

# INES AITSAHALIA

PHD CANDIDATE

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NEW YORK CITY, NY

## EDUCATION

2022-Present **PhD, Columbia University**, Center for Theoretical Neuroscience (est. 2027)

**MPhil** in Neurobiology and Behavior (2025)

**MA** in Neurobiology and Behavior (2024)

2018-2022 **Princeton University**

BA in Neuroscience, Cognitive Science

*Graduated with Honors & John Brinster, Class of 1943, Senior Thesis Prize*

## RESEARCH

2024-Present **Iigaya Lab & Stachenfeld Lab**, Columbia University

PhD Candidate

2025 **BASIS AI Research Institute**,

PhD Research Intern

2023 **Miller Lab**, Center for Theoretical Neuroscience at Columbia University

PhD Student Researcher (Rotation)

2022 **Horga Lab**, Columbia University and New York State Psychiatric Institute

PhD Student Researcher(Rotation)

2022 **Iigaya Lab**, Columbia University and New York State Psychiatric Institute

PhD Student Researcher (Rotation)

2020-2022 **Niv Lab**, Princeton University

Undergraduate Research Assistant

Summer Research Intern

## PUBLICATIONS & PRESENTATIONS

### **Hierarchical Latent Structure Learning through Online Inference**

**I. Aitsahalia** & K. Iigaya

2026, *Neural Computation* (in revision)

“Baseline Negative Symptoms Predict Medication-Specific Adverse Events and Early Treatment Discontinuation in Schizophrenia”

S. Kim, **I. Aitsahalia**, B.H. Yalcinkaya-Selimbeyoglu, M. Yue, Z. Zhang, G. Horga, S. Stroup, S. Chen, K. Iigaya

2026,  
Poster at Society of Biological Psychiatry (SOBP)

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Striatal acetylcholine enables latent-state creation during reversal learning</b><br>D. Lowes, D. Urey, S. Fernandez, <b>I. Aitsahalia</b> , S. Ennis, M. Prado, Kiyohito Iigaya, Christoph Kellendonk | 2026, BioRxiv                                                |
| <b>Multimodal Individual Differences in Human Decisions Can Arise from Distinct Cognitive Graphs</b><br>P. Sepulveda, A. Gungi, <b>I. Aitsahalia</b> , K. Iigaya                                          | 2026, BioRxiv                                                |
| <b>Building Goal-Directed Cognitive Graphs</b><br>A. Gungi, P. Sepulveda, <b>I. Aitsahalia</b> , Marta Blanco-Pozo, K. Iigaya                                                                             | 2026, BioRxiv                                                |
| <b>Inferring cognitive strategies from groups of animals in natural environments</b><br><b>I. Aitsahalia</b> , T. Botch, S. Gu, T. O'Connell, R. Seigel, R. Peterson, D. Batenkov, E. Mackevicius         | 2025<br>NeurIPS workshop paper                               |
| <b>Anticipatory value computation as a new transdiagnostic target</b><br>P. Sepulveda, <b>I. Aitsahalia</b> , K. Kumar, T. Atkin, K. Iigaya                                                               | 2024, Biological Psychiatry                                  |
| <b>Computational Psychiatry and Eating Disorders</b><br><b>I. Aitsahalia*</b> , P. Sepulveda*, S. Dong*, K. Iigaya (*denotes co-first authors)                                                            | 2024,<br>Oxford Handbook on Eating Disorders                 |
| <b>"Learning from Outliers"</b><br><b>I. Aitsahalia</b>                                                                                                                                                   | 2025<br>Talk at NY Memory Hub                                |
| <b>"A multi-level latent cause framework for the study of delusions"</b><br><b>I. Aitsahalia</b> , G. Horga, K. Iigaya                                                                                    | 2025,<br>Poster at Computational Psychiatry Conference (CPC) |
| <b>Building Deterministic Causal Graphs Using Reinforcement Learning in Cognitive Tasks with Evolutionary Bias</b><br><b>I. Aitsahalia</b> , A. Gungi, P. Sepulveda, K. Iigaya                            | 2024, Computational Cognitive Neuroscience                   |
| <b>"Unified anticipation model can explain information-seeking around negative outcomes"</b><br><b>I. Aitsahalia</b> , P. Sepulveda, V. Fisher, S. Dong, K. Iigaya                                        | 2024,<br>Poster at CPC                                       |
| <b>"Common cause inference as a generative model of delusions"</b><br>M. Schaffer, A. Veliksvkaya, <b>I. Aitsahalia</b> , K. Iigaya, G. Horga 2024 poster                                                 | 2024,<br>Poster at CPC                                       |
| <b>"Bridging reinforcement learning and causal inference: Emergence of grid cells in abstract tasks"</b><br>A. Gungi, P. Sepulveda, <b>I. Aitsahalia</b> , M. Schaffer, K. Iigaya                         | 2024,<br>Poster at CoSyNe                                    |
| <b>"Information seeking around aversive outcome anticipation"</b><br><b>I. Aitsahalia</b> , P. Sepulveda, S. Dong, R. Chen, Y. Chai, K. Iigaya                                                            | 2023,<br>Poster at SfN                                       |
| <b>"Computation of Anticipatory Value as a New Transdiagnostic Target"</b><br><b>I. Aitsahalia</b> , P. Sepulveda, K. Kumar, K. Iigaya                                                                    | 2023,<br>Poster at CPC                                       |
| <b>"Alterations in Perceptual Learning Drive Speech Hallucinations in Medication-Free Patients with Schizophrenia"</b><br>J. Buck, <b>I. Aitsahalia</b> , D. Serrano, G. Horga                            | 2023,<br>Poster at CPC                                       |
| <b>"Why do some beliefs and behaviors resist updating?"</b><br>S. Pisupati, <b>I. Aitsahalia</b> , S.N. Quinard, I.M. Berwian, A.J. Langdon, Y. Niv                                                       | 2023,<br>Poster at CPC                                       |
| <b>"Latent-cause inference under varying beliefs about</b>                                                                                                                                                | 2022,                                                        |

randomness captures individual differences in fear extinction”

Poster at SfN

I. Aitsahalia, S. Pisupati, S.N. Quinard, I.M. Berwian, Y. Niv

“Over- and under-segmentation of latent causes lead to different types of extinction failures”

2021,  
Poster at SfN

I. Aitsahalia, S. Pisupati, S.N. Quinard, I.M. Berwian, Y. Niv

## TEACHING & SERVICE

2025 **Guest Lecturer (Biostatistics)** Columbia University

2025 **Committee on Rigor, Neurobiology and Behavior** Columbia University

2024 **Preventing and Managing PTSD, certification**

2023, 2024, 2025 **Course Instructor, Math Tools for Theoretical Neuroscience**

Columbia University

2023-2024 **Course Instructor, Mind/Brain Disorders** Barnard College

2023, 2025 **Teaching Assistant, Analysis for Neuroscientists** Columbia University

2023-2025 **Teaching Assistant, Introduction to Cognitive Science** Barnard College

2022-Present **Women on Two Floors** women in computational neuroscience at Columbia

2021-2022 **Subcommittee on Inclusive Teaching** Princeton University

2022-Present **Dragonfly Mental Health**

2021-2022 **Empowering Diversity and Promoting Scientific Equity at PNI** Princeton University

## AWARDS

2026 **Thinking about Thinking Ambassador** Thinking about Thinking

2026 **Junior Scientist in Theoretical Neuroscience Fellow** Janiela Research Campus

2026 **CTCN Next Generation Fellowship** Washington University

2025 **Best Data Blitz** New York Memory Hub Conference

2025 **Student Leadership and Service Award** Columbia University

2025 **Presidential Teaching Award Finalist** Columbia University

2024-2025 **Lead Teaching Fellow**, Columbia University Center for Teaching and Learning

2022-Present **NSF Graduate Research Fellow**, Cognitive Science

2022 **Knox Studentship**, Cambridge University *Offer declined*

2022 **John Brinster, Class of 1943, Senior Thesis Prize**, Princeton University

2021 **Nancy J. Newman, MD '78 & Valerie Biousse, MD Research Fund for Neuroscience**,  
Princeton University

2021 **Program in Cognitive Science Summer Research Award**, Princeton University